

Priya Nair

Senior Machine Learning Engineer

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Professional Summary

Senior ML engineer with 8 years of experience shipping Python-based LLM and NLP systems. Deep expertise in QLoRA fine-tuning, Hugging Face Transformers, and production FastAPI inference services on consumer and cloud GPUs. Passionate about structured generation, evaluation harnesses, and document understanding pipelines.

Experience

Senior Machine Learning Engineer — Helix Document AI (2021 – Present)

- Led QLoRA fine-tuning of Llama 3.1 8B for structured JSON extraction from resumes and contracts; cut GPU cost ~4× vs full FT.
- Built PDF parsing + chunking stack with PyMuPDF and custom PII redaction; processed 2M+ pages/month.
- Designed FastAPI + vLLM inference service with p95 latency <400ms on A10G; exported GGUF builds for laptop demos.
- Owned evaluation suite (precision/recall, calibration, human agreement) and experiment tracking in Weights & Biases.

Machine Learning Engineer — Northstar Analytics (2018 – 2021)

- Fine-tuned BERT/RoBERTa models for multi-label document classification; improved F1 by 12 points over baseline.
- Implemented training pipelines with PyTorch, PEFT, and accelerate on multi-GPU nodes.
- Deployed model endpoints on AWS (SageMaker + ECS) with Docker and CI via GitHub Actions.

Software Engineer (Python) — CloudLedger Inc. (2016 – 2018)

- Built backend microservices in Python (Flask/FastAPI precursors) for financial document ingestion.
- Wrote unit/integration tests (pytest) and enforced linting standards.

Skills

Python, PyTorch, Hugging Face Transformers, PEFT, bitsandbytes, Unsloth, QLoRA, TRL, vLLM, llama.cpp/GGUF, FastAPI, PyMuPDF, Docker, AWS, W&B, pytest, ruff, Git

Education

- M.S. Computer Science — Stanford University, 2016
- B.S. Computer Science — UC San Diego, 2014

Projects & Portfolio

- Open-source resume-scorer demo: Llama-3.1-8B QLoRA + GGUF export (GitHub: example/priya-resume-scorer).
- Blog series on structured JSON generation and schema-constrained decoding.